

Cross-Platform Reproduction of the QDE Benchmark

1 Cross-Platform Reproduction: Windows/x86-64 with OpenBLAS

The benchmark reproduces bit-identically on the validated platform (Apple silicon, Python 3.9, Accelerate BLAS). To establish which results are properties of the method rather than of that machine’s floating-point environment, the full pipeline was re-executed end to end on an unrelated platform: Windows on x86-64 with OpenBLAS and a four-minor-version-newer Python. This section compares the two sets of results.

1.1 Execution environment

	Original	Reproduction
Platform	macOS 26.5.1, arm64	Windows 11 (10.0.26200), x86-64
CPU / memory	Apple silicon	12 cores, 15.9 GB
Python	3.9.6	3.12.10
BLAS / LAPACK	Accelerate	scipy-openblas 0.3.33
NumPy	2.0.2	2.5.1
SciPy	1.13.1	1.18.0
scikit-learn	1.6.1	1.9.0
PyTorch	2.8.0	2.13.0+cpu
pandas	2.3.3	2.3.3
statsmodels	0.14.6	0.14.6
XGBoost	2.1.4	2.1.4
LightGBM	4.6.0	4.7.0
reservoirpy	0.4.2	0.4.2
matplotlib	3.9.4	3.11.1

Table 1: Execution environments. All dependencies remain within the version caps declared in `pyproject.toml`, but the exact lockfile was deliberately not pinned, so the comparison spans both a different BLAS implementation and a drifted dependency closure.

The pipeline was run from a clean tree with no cached quantum features, so every value below was regenerated from source and seeds. Wall-clock time for the full sequence was 27.7 h (99,689 s) on CPU.

1.2 Results that reproduce exactly

Table 2 compares the two runs on every quantity the thesis reports as a finding. All agree.

Two entries deserve emphasis. First, the qubit-scaling family preserves not only the 213/1 split but the *identity* of the single reversal among 214 pre-specified per-seed comparisons: it falls in the same cell (Hénon, $n = 4$, QRC versus ELM(F)) on both platforms. Second, the Mackey–Glass matched-budget NRMSE agrees to all seven retained significant figures, a relative difference of 1.1×10^{-7} .

Quantity	Original	Reproduction
Pre-specified family size	214	214
QRC significantly worse	213	213
QRC significantly better	1	1
Identity of the reversal	Hénon $n=4$ vs. ELM(F)	Hénon $n=4$ vs. ELM(F)
Core family (excluding $n=12$)	210	210
All survive Holm–Bonferroni	yes	yes
Median error ratio vs. NG-RC	7.3 [6.2, 9.4]	7.3 [6.2, 9.4]
Median error ratio vs. ELM(F)	4.6 [3.0, 5.2]	4.7 [2.9, 5.4]
Median paired difference (ELM(F) / NG-RC)	3.5e-3 / 4.4e-3	3.5e-3 / 4.4e-3
Selected NG-RC, Hénon	$k=3, d=2$	$k=3, d=2$
Selected NG-RC, Lorenz-63	$k=8, d=3$	$k=8, d=3$
Selected NG-RC, Mackey–Glass	$k=8, d=3$	$k=8, d=3$
Matched-budget verdicts (v4 / v6 / rich)	F / F / T	F / F / T
Selected scaler (all three arms)	minmax	minmax
Mackey–Glass <code>qrc_rich</code> NRMSE	4.618213e-3	4.618213e-3

Table 2: Quantities that agree across platforms. The ELM(F) error ratio differs only in the third significant figure, with overlapping bootstrap intervals; every other entry is identical.

The headline conclusion is therefore unchanged on the second platform: no evaluated QRC configuration outperformed the strongest classical comparator, and 213 of 214 pre-specified comparisons show the quantum model significantly worse, all surviving multiplicity correction.

1.3 Results that differ

Table 3 lists every quantity that moved. Each is either measured on Hénon or is a p -value.

Quantity	Original	Reproduction	Rel. diff.
<code>qrc_rich</code> NRMSE (quantum)	1.110878e-3	1.120515e-3	0.87 %
<code>qrc_v6</code> NRMSE (matched ESN)	5.217691e-2	5.159655e-2	1.11 %
<code>qrc_v4</code> NRMSE (quantum)	2.975403e-1	2.911201e-1	2.16 %
<code>qrc_v4</code> NRMSE (matched ESN)	7.484244e-2	7.224951e-2	3.46 %
<code>qrc_v6</code> NRMSE (quantum)	2.706064e-1	2.575529e-1	4.82 %
<code>qrc_rich</code> NRMSE (matched ESN)	2.003425e-2	1.904232e-2	4.95 %
Maximum raw p (scaling family)	5.6e-5	4.77e-4	8.5×
Maximum Holm-adjusted p	9.7e-5	4.77e-4	4.9×

Table 3: Quantities that differ across platforms. No decision changes: the largest adjusted p -value remains two orders of magnitude below the $\alpha = 0.05$ threshold.

This is the pattern the artifact’s numerical analysis predicts. Lorenz-63 and Mackey–Glass are produced by integrators that agree bit-identically across environments, whereas Hénon is an iterated map: its trajectory agrees exactly for roughly fifty steps, then differs in the last place, and the positive Lyapunov exponent amplifies that difference to order unity by approximately step 134. Every Hénon-derived quantity is therefore environment-specific at the scale of the signal. The p -values, being functions of the forecast residuals, are the most numerically sensitive derived statistic in the study; the largest adjusted value grew by roughly a factor of five and remained far inside the rejection region.

The contrast between the two systems is itself the confirming evidence. On the same run, on the same BLAS, the Mackey–Glass matched-budget value reproduces to seven significant figures while the corresponding Hénon values move by up to 5 %. This localises the divergence to the

data-generating process rather than to the modelling or evaluation pipeline: were the pipeline itself environment-sensitive, the flow systems would drift too.

1.4 Numerical notes

Two observations qualify the comparison.

Re-executing a single QRC cell on the same machine reproduced the recorded NRMSE to approximately 33 units in the last place ($\sim 3 \times 10^{-17}$ absolute), attributable to nondeterministic reduction order in the multithreaded BLAS used by the ridge solve. Bit-identity should therefore be claimed for the validated Apple silicon platform specifically, not as a general property of the pipeline.

Two changes to the artifact source were required before the pipeline would run to completion on Windows: explicit UTF-8 encoding for text output, and a memory-efficient representation of the quantum readout operators, whose dense form requires approximately 12 GiB at $n = 12$ and is impracticable on a 16 GiB host. The readout substitution is an exact factorisation of the same Pauli observables—each is a Pauli string, so the operator acts as a diagonal scaling composed with an index permutation—and was verified bit-identical to the dense operator per observable and end to end at the level of the reported NRMSE. No value in either table is affected.

1.5 Conclusion

Every claim the thesis makes about ordering, model selection and statistical significance survives transplantation to an unrelated floating-point environment. The quantities that moved are confined to Hénnon-derived values and to p -values, both identified in advance as environment-sensitive, and none of the movement changes a decision. The results are properties of the method, not of the machine on which they were first produced.